INCH-POUND

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29 March 1991
SUPERSEDING
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25 April 1990

MILITARY SPECIFICATION

SOLID STATE X-BAND SOURCE: 10548182

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Solid State X-Band Source, 10548182, herein referred to as the X-band source, a subassembly of the Receiver-Transmitter, RT-860/VPS-2, a part of Radar Set AN/VPS-2 and a subassembly of MX-8046A/TPM-22, a part of Radar Test Set AN/TPM-22 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and DispositiOn of
MIL-R-50651 - Radar Set AN/VPS-2, Range Only Radar

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1285

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance **Terms and Definitions**
MIL-STD-810 - Environmental Test Methods **and Engineering Guidelines**
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation. ,

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548182 - Solid State X-Band Source
10549175 - Oscillator, Module
11822700 - Noise Tester Assembly

PACKAGING DATA SHEET

PS10548182 - packaging of X-Band Source

(Copies of **drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)**

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing In this specification, however, **shall supersede applicable laws and regulations unless a specific exemption has been obtained.**

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 General provisions.

3.2.1 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548182, all associated drawings, and with all requirements specified in applicable specifications *and* standards. Where-applicable, the X-band source shall meet the requirements of MIL-F-13926 as follows: .

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.2.2 **Interchangeability.** The X-band source shall be designed so that all sources are compatible with all crystal oscillators (per drawing 10549175). The driver section, coaxial multiplier, and waveguide multiplier shall be electrically and mechanically interchangeable from unit to unit.

3.3 Environmental service conditions.

3.3.1 Vibration. The X-band source shall not suffer damage and shall meet the requirements of 3.5 (dc voltage and current only), 3.5.1, 3.5.2, 3.5.3, and 3.5.6.2 after exposure to sinusoidal vibrations, in each of three vertically perpendicular axes with one axis normal to the vertical curve of the item as follows:

Frequency Range	Amplitude (one half Total Excursion	Acceleration	Time (Each Axis)
5-10 Hz	.060 inch		1 Hr
11-20 Hz	.030 inch		2 Hrs
21-400 Hz		9.OG	2 Hrs

3.3.2 Shock. The X-band source shall not suffer damage and shall meet the requirements of 3.5 (dc voltage and current only), 3.5.1, 3.5.2, 3.5.3, and 3.5.6.2 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 High shock. The X-band source shall meet the requirements of 3.5 (dc voltage and current only), 3.5.1, 3.5.2, 3.5.3, 3.5.6.2 and after exposure to a total of 18 impulse shocks consisting of one shock impulse of 75 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

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3.3.4 Altitude.

3.3.4.1 0perating The X-band source shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4.2 Non-operating. The X-band source shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.5 Temperature.

3.3.5.1 OPerating. The X-band source shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between the constant temperature extremes of -40 degrees F and +155 degrees F.

3.3.5.2 Non-operating (storage). The X-band source shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.6 Humidity (non-operating) The X-band source shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram;

3.5 Performance.

3.5.1 Operation. The X-band source shall perform within specifications when operating from a dc power supply with these characteristics:

28 Vdc regulated to 1%.
Ripple maximum of 0.01% peak-to-peak.
Maximum output current 1A.

3.5.1.1 The X-band source shall be internally protected against damage resulting from the application of primary dc power of the incorrect polarity.

3.5.2 Output frequency. The output frequency range shall be from 9090 to 9190 MHz inclusive. The specific operating frequency shall be determined by choice of an appropriate oscillator.

3.5.3 Output power. The output power requirements shall be met when terminated by an impedance mismatch with a VSWR of 1.25: 1 to 1.3:1 at any arbitrary phase angle.

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Condition A - The output power shall be at least 95 mW, but *no* greater than 115 mW when **the** input voltage is 28.5 plus or minus 1 percent Vdc across the frequency band of 9090 MHz to 9190 MHZ inclusive on all delivered items.

Condition B - The output power shall be at least 80 mW, but no greater than 120 mW **when the input voltage is 28.5 plus or minus 1 percent Vdc when testing to or after exposure to the requirements of paragraph 3*3.**

Upon satisfactory completion of testing or after exposure to requirements of paragraph 3.3, output power readjustment is permitted in order to meet the requirements of condition A.

3.5.4 Long term stability. The fractional frequency deviation during a 24 hour period shall not exceed:

$$\frac{(\text{delta})\ f}{f} = 1 \text{ part per } (10^4)$$

where:

(delta) F is defined as the standard deviation of f and f_o. This long range stability must be achieved when the temperature, rf load, and power supply voltages vary simultaneously throughout their permitted operating ranges as specified herein. The long term stability shall be compatible with the requirements of oscillator module drawing 10549175. Measurement methods and characteristics of measuring equipment must be clearly given by the vendor and approved.

3.5.5 **Warm-up time. The X-band source shall meet the requirements of 3.4 after a 90 second warm-up time.**

3.5.6 Spectral Purity. Extraneous power components in the output comprise two categories broadband and narrow band 1/: (1) The broadband category is noise which stems from instability of the crystal source and from inherent noise in other semiconductor devices (2) the narrow band are specular **signals which result from non-linear processing (multiplication) of the basic frequency and from any heterodyning which may be employed.**

This measurement sha11 be taken with the **oper**sting volts**ge** set at 29.5 plus or minus .5 Vdc for all spectral **purity** requirements. The frequency band is esttablished by the s'**ix plug**-in modules of drawing 10549175.

1/ Narrowband implies a cw constant frequency signal with spectral bandwith less than 300 Hz.

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3.5.6.1 **Power source noise spectral density (broadband).** Power source noise power spectral density $N_o(f)$ in watts/Hz, shall be such that the integrated power of $N_o(f)$ over the range from 600 Hz to 2 MHz when weighted by a factor of $2(1-\cos 2(\text{pi})\ ft_d)$ shall be at least 60 dB below the total power, P_Q.

$$2\int_{600\ Hz}^{2\times10^6\ Hz}(1-\cos 2\pi f t_d)\ N_o(f-fo)\ df \leq 10\text{-}6P_o \quad (1)$$

The following approximateion may be used for (l):

$$2\left[4\pi^2(10^{-9})\ f^2 \cdot \int_{600\ Hz}^{10^4\ Hz}(f-f_o)\ df + N_{o_{10^4}}\int_{10^4\ Hz}^{2\times10^6\ Hz}(f-fo)\ df\right] \leq 10\text{-}6P_o$$

The noise power spectral denisity shall be measured utilizing nelse measuring equipment and the results converted to units of watts per hertz. The method of measurements and characters of the measuring equipment must be clearly given by the vendor.

Under conditions of vibration and shock, as outlined in 3.3.1 and 3.3.2 respectively, the integrated noise power as computed by equation (1) above should not exceed $2 \times 10^{-6} P$; which includes an additional contribution of 3 dB to the noise power spectral density of the source under these conditions.

Parasitic oscillations are undesirsble and shall be eliminated through design and alignment. Any vestiges of parasitic oscillations remaining must be included within the noise power spectral density limits.

Figure 1 shows the curve of power spectral density requirements. No discontinuities or spikes shall be present on any portion of the Curve shown.

3.5.6.2 Specular output (narrowband). Harmonics, sub-harmonics, image frequencies or other components resulting from the development of the x-band output from the crystal frequency shall be 45 dB or more below the desired X-band output except as modified at low frequencies by 3.5.6.1. No crystal oscillator moding which results in the generation of extraneous output shall be allowed.

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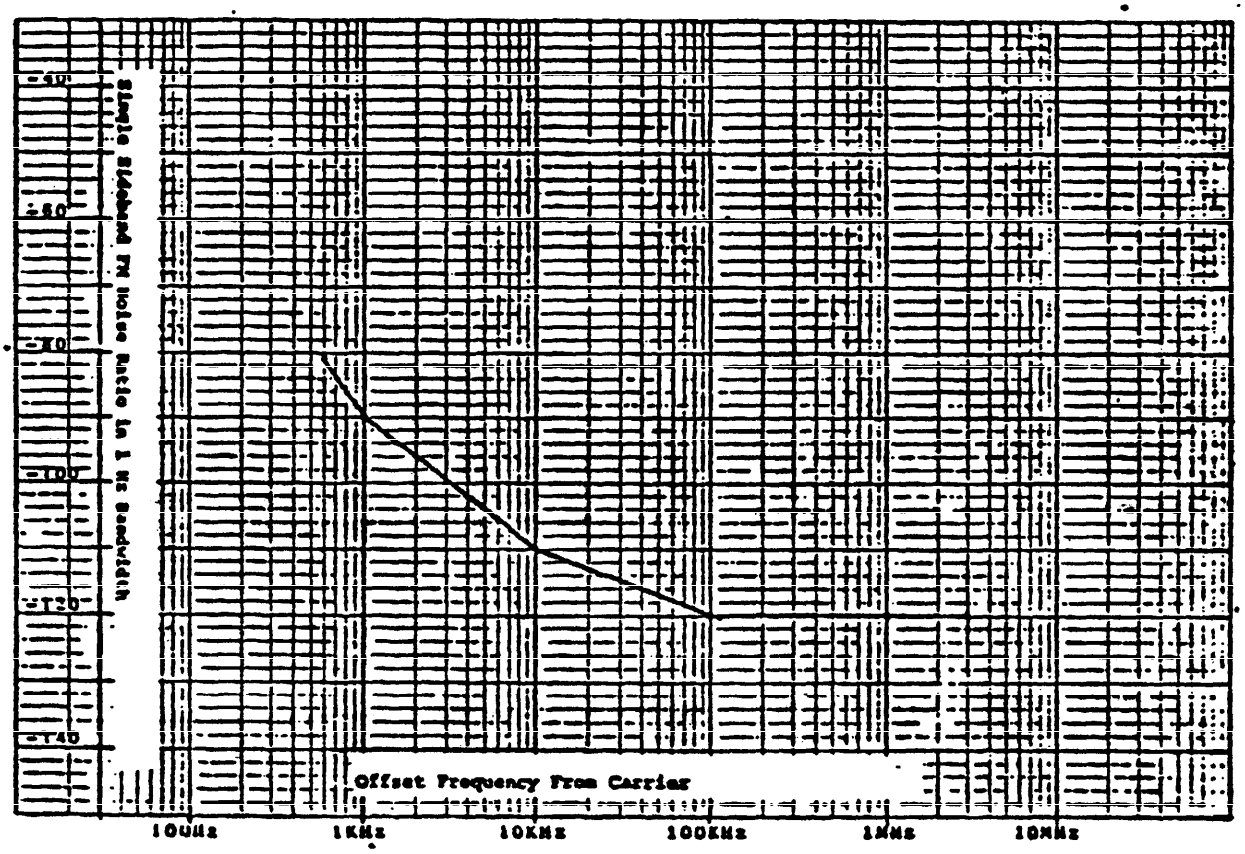


FIGURE 1. Power spectral density curve.

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3.5.7 **System compatibility.** It must be clearly demonstrated that the X-band source be compatible with the **AN/VPS-2 ROR Radar** Specification **MIL-R-50651.** The X-band source shall meet the following requirements of MIL-R-50651:

a. Transmission characteristics (Paragraph 3.5.6)
b. System noise figure (Paragraph 3.5.8)
c. Subclutter visibility (Paragraph 3.5.9.8)

3.6 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

3.7 Burn-in. After the x-band source stabilizes at temperatures of -42 ± 2 degrees C, and + 73 ± 2 degrees C it shall soak for 30 minutes at each temperature. It shall then be returned to standard ambient temperature before meeting the requirements of 3.5.

3.8 First article. When specified in the contract or purchase order, **a** sample shall **be subjected to first article inspec**tion in accordan**ce with the technical provisions herein (see 4.3).**

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4*1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's **overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance** comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	**Temperature**	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The requirement for first article and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract and the applicable drawing. The sample for the first article tests shall consist of three x-band sources manufactured in the same manner, using the same materials, equipment, processes and procedure as in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.3.1.1 Government testing. When the Government is responsible for conducting first article tests, the contractor, prior to submitting the sample to the Government, shall Inspect the sample to ensure that it conforms to all the requirements of the contract and the applicable drawings. The contractor shall submit a record of this inspection with the sample including certified test report (CTR) or certificate of conformance (COC) for the materials.

4.3.1.2 Contractor testing. When the Contractor is responsible for conducting first article tests, the sample shall be inspected by the contractor for all the requirements of the contract, all of the examinations and tests specified in Table I (Functional Test) and Table II (Special Sampling) herein and **to all** requirements of the applicable drawings. The sample and a **record** of this inspection, including CTR/COC, shall be submitted **for the** Government approval. The Government reserves the right to w**itness** the Contractor's inspection.

4.3.2 **Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of** any assembly, component **or test specimen to** comply with **any of the stated requirements.**

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4.4 **Quality conformance inspection.**

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior *or* subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be **performed on a single defect (individual characteristic) basis in accordance the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.**

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

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TABLE I. Functional tests
(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
101.	Electrical	3.4	4.7.2
102.	Performance	3.5	4.7.3
103.	Burn-in	3.7	4.9
Minor:		None	N/A

4.5 Special sampling. One X-band source out of the first 50 produced and one out of each 100 Produced thereafter shall be selected at random. The sample shall have-met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract the Items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
104.	Vibration	3.3.1	4.7.1.1
105.	Shock	3.3.2	4.7.1.2
106.	High Shock	3.3.3	4.7.1.3
107.	Altitude	3.3.4	4.7.1.4
108.	Temperature	3.3.5	4.7.1.5
109.	Humidity	3.3.6	4.7.1.6
Minor:		None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in 'Examination" and 'Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III shall be used in the **performance of the applicable test methods and procedures as specified in 4.7.**

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TABLE III. Test equipment list

ITEM NUMBER	DESCRIPTION
1. **Power supply**	**Type: laboratory supply.** **Minimum requirements: 20 to 30 Vdc.** Load **regulation: 0.01%** **Load current capability: 1.5A.** **Ripple and noise: lmV** rms or 5mV Peak-to-peak **maximum from dc to 20 MHZ.**
2. Power supply	**Type: laboratory supply.** **Minimum requirements: O to 10 Vdc.** **Load regulation:** 0.02%. Load current capability: **100mA.** **Ripple and noise: lmV rms to** 5mV peak-to-peak maximum from dc to 20 MHZ.
3. **Ammeter, dc**	**0-1.5 A.** Accuracy: 2%
4, 5, 6, 7, 8. Attenuator	Frequency range: 9150 **to** 9250 **MHz.** Attenuation: 20 dB maximum. Average power: 2W minimum. Peak power: 2kW minimum. Max. VSWR : 1.0:1.15.
9. Frequency meter	Type: direct reading resonant cavity. Frequency range: **9150 to 9250 MHz maximum.** **Overall accuracy: 0.08%.**
10. Directional coupler (10 dB)	Bandwidth: **9150 to 9250 MHz minimum.** **Peak power rating: 2000 W minimum.** **Coupling: 10 dB ± 0.4 dB.** **Mainline VSWR: 1.0:1.1 maximum.** **Secondary line VSWR: 1.0:1.2 maximum.**
11. Directional coupler (3 dB)	Bandwidth: 9150 **to** 9250 MHz minimum. Peak power rating: 2000 W minimum. Coupling: 3 dB ± 0.4 dB. Mainline VSWR: 1.0:1.1 **maximum.** Secondary line VSWR: 1.0:1.2 **maximum.**
12. Thermistor mount (compatible with item 13)	Frequency range: 9150 to 9250 MHz minimum. Maximum **average input rf power: 10mW minimum.**

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TABLE 111. Test equipment list (cent.)

ITEM NUMBER	DESCRIPTION
13. Power meter (compatible with with 12)	Frequency range: 9150 to 9250 MHz minimum. Power range: lmw full scale. Accuracy: ± 1% of full scale. Type : Thermistor sensing element.
14, 15, 16. Isolator	Frequency range: 9150 to 9250 MHz minimum. Isolation: 30 dB minimum. Insertion: 1 dB maximum. VSWR: 1.0:1.15 maximum. Average power: 3 W minimum.
17. Digital voltmeter	Range: 0-100 Vdc minimum. Minimum accuracy: 0.1% of reading + 0.1% of scale.
18. Transmission cavity	Frequency range: 9150 9250 MHz minimum. Q: 16,000 minimum.
19. Rf switch	Frequency range: 9150 to 9250 MHz minimum. Insertion: 0.1 dB maximum. Crosstalk: - 60 dB minimum. VSWR: 1.0:1.10 maximum. Single-pole, double-throw. Average power: 3 W minimum.
20. Termination	Frequency range: 9150 to 9250 MHz minimum. VSWR: 1.0:1.05 minimum. Average power: 3 W minimum. Peak power: 600 W minimum.
21. Fixed mismatch	Frequency range: 9150 to 9250 MHz minimum. VSWR: 1.0:1.3. Average power: 3 W minimum. Peak power: 600 W minimum.
22. phase shifter	Frequency range: 9150 to 9250 MHz minimum. Phase shift: Variable 0-90 degree.

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TABLE 111. Test equipment list (cent.)

ITEM NUMBER	DESCRIPTION
23. Hybrid t e e minimum.	Frequency range: 9150 to 9250 MHz VSWR : 1.0:1.15 Max. E, H and side arms. Power division: 0.25 dB maximum Isolation: 40 dB minimum.
24. DC block	Frequency range: 9150 to 9250 MHz minimum. VSWR : 1.0:1.15 minimum Insertion: 0.1 dB maximum.
25, 26. Matched crystal detectors	Frequency range: 9150 to 9250 MHz minimum. VSWR : 1.0:1.5 minimum. Response flatness: ± 0.25 dB. Matching: ± 0.3 dB for square law application up to 75 mV output
27. 60 Hz filter	L=12mH + 10% c = 6.8f ± 10%
28, 29. Low noise amplifiers	Combined frequency response spread: 600 Hz to 2 MHz. Input impedance: 1 megohm minimum. Output impedance: 600 ohms maximum Noise referred to input: 15 V maximum. Gain: 500.
30, 31. Tunable Wave analyzer	Combined frequency response spread: 600 Hz to 2 MHz minimum. Bandwidth: 3 dB 200 Hz minimum. Amplitude accuracy: ± 6% maximum. Input impedance: 60 ohms to 1 megohm. Input voltage range: 10 v to 100 v minimum.
32. Not Used.	

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TABLE 111. Test equipment list (cent.)

ITEM NUMBER	DESCRIPTION
33, 34, 35, 36, 37, 38. Oscillator modules	Drawing: 10549175
39. Counter	Counter intervals: 10^{-4} to 1 second minimum. Start and stop trigger sensitivity: 0.1 V to 100 V minimum.
40. Signal generator	Frequency range: 9150 to 9250 MHz minimum. Rf output: 0 dBm to -127 dBm. Modulation: External fm, 600 Hz to 2 MHz minimum, 20V/MHz minimum. Frequency stability: .006%/C° maximum, .02% maximum for ± 10% line voltage change
41. Oscillator	Frequency range: 600 Hz to 2.5 MHz minimum. output : 0 to 10 Vat.
42. Spectrum analyzer	Frequency range: 9150 to 9250 MHz minimum. Resolution: 3 dB bandwidth selectable from 100 Hz to 3 KHz. Log display: 70 dB minimum.
43. Diode bias/switch-	Circuitry for biasing mixer diodes ing (special) both in parellel and separate and metering bias voltage.
44, Noise Tester Part No. 11822700	Special Noise Tester specifically designed to test to X-band source noise. Alternate equipment, See paragraph 4.7.3.6, not available as GFE .

4.7 Methods of inspection.

4.7*1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

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4.7.1.1 vibration. The test shall be performed as specified in Method 514, Category 8, Procedure 1, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The X-band source shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the X-band source shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 shock. This test shall be performed as specified in Procedure I, section II of Method 516 of MIL-STD-810 except that:

***a.* The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.**

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

***c.* The maximum g shall occur at 5.5 milliseconds ± 10%.**

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz. After the application of all the shock impulses, the X-band source shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 High shock. This test shall be performed as specified in procedure I, Section II of Method 516 of MIL-STD-810 except that:

***a.* The post test functional check (Step 5 of Procedure I) need not** be conducted until completion of all shock-impulses.

b. The shock pulses shall be 75 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within + 10% from 20 to 2,000 hertz. After the application of all the shock—impulses, the X-band shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.3.

4.7.1.4 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810 . The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.5 Temperature.

4.7.1.5.1 High (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.5. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.1.5.2 LOW (operating and non-operating) This test shall be performed as specified in procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.1.6 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.6.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. The X-band source shall be electrically tested using the test equipment configuration shown in Figure 2 and 3 or an approved equal.

4.7.3.1 Dc input power required. Assemble test equipment as shown in Figure 2. verify compliance with 3.5.1.

4.7.3.1.1 procedure. Determine current drawn by X-band source. Meter shall indicate less than 1 Adc.

4.7.3.1.2 Reverse power input leads to x-band source. observe that current meter remains at zero indicating no conduction when polarity is reversed from normal.

4.7.3.2 Output frequency. With test equipment as shown in Figure 2, verify compliance with 3.5.2.

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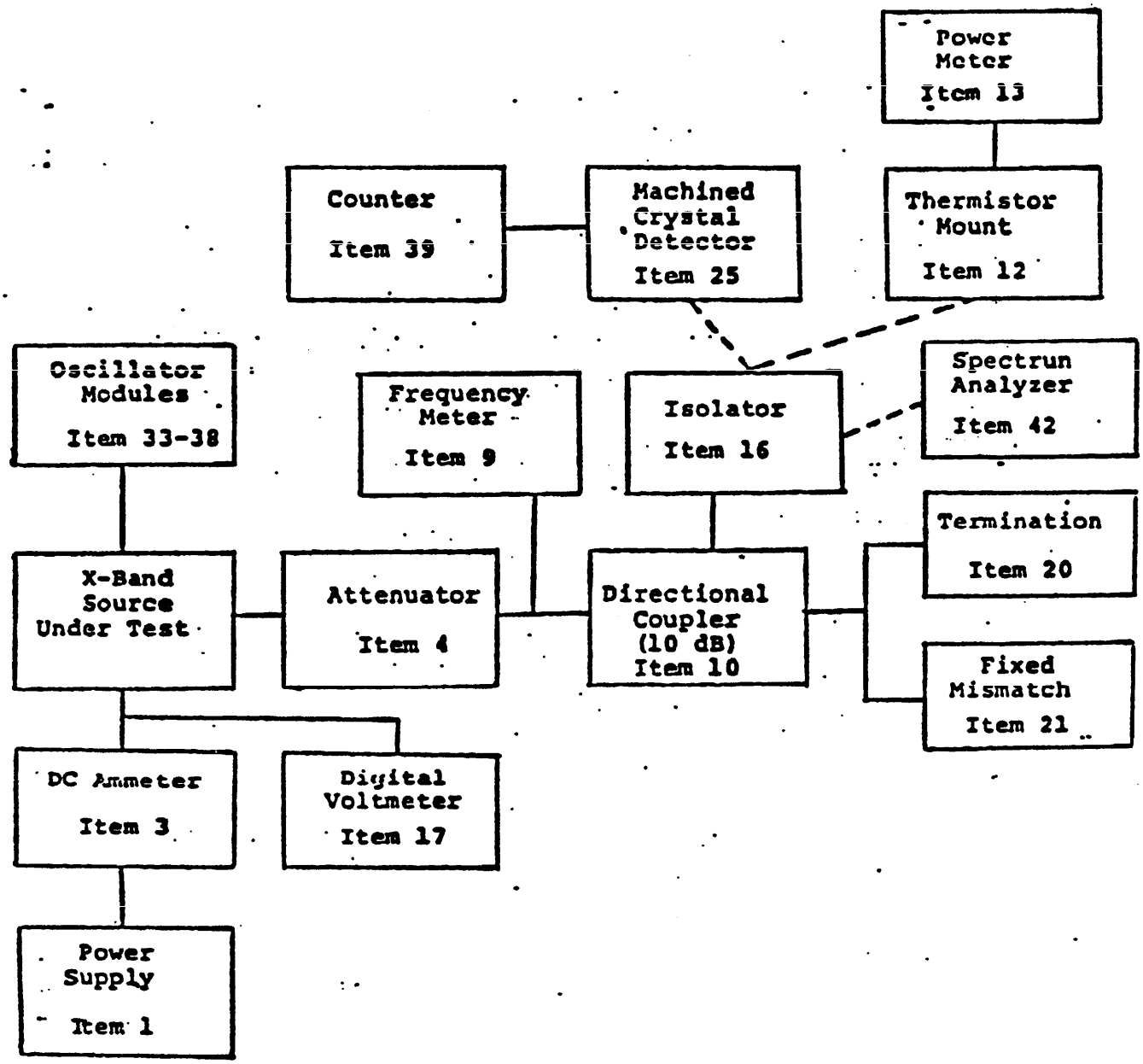


FIGURE 2. Tuning, frequency, output power and start-up time set up.

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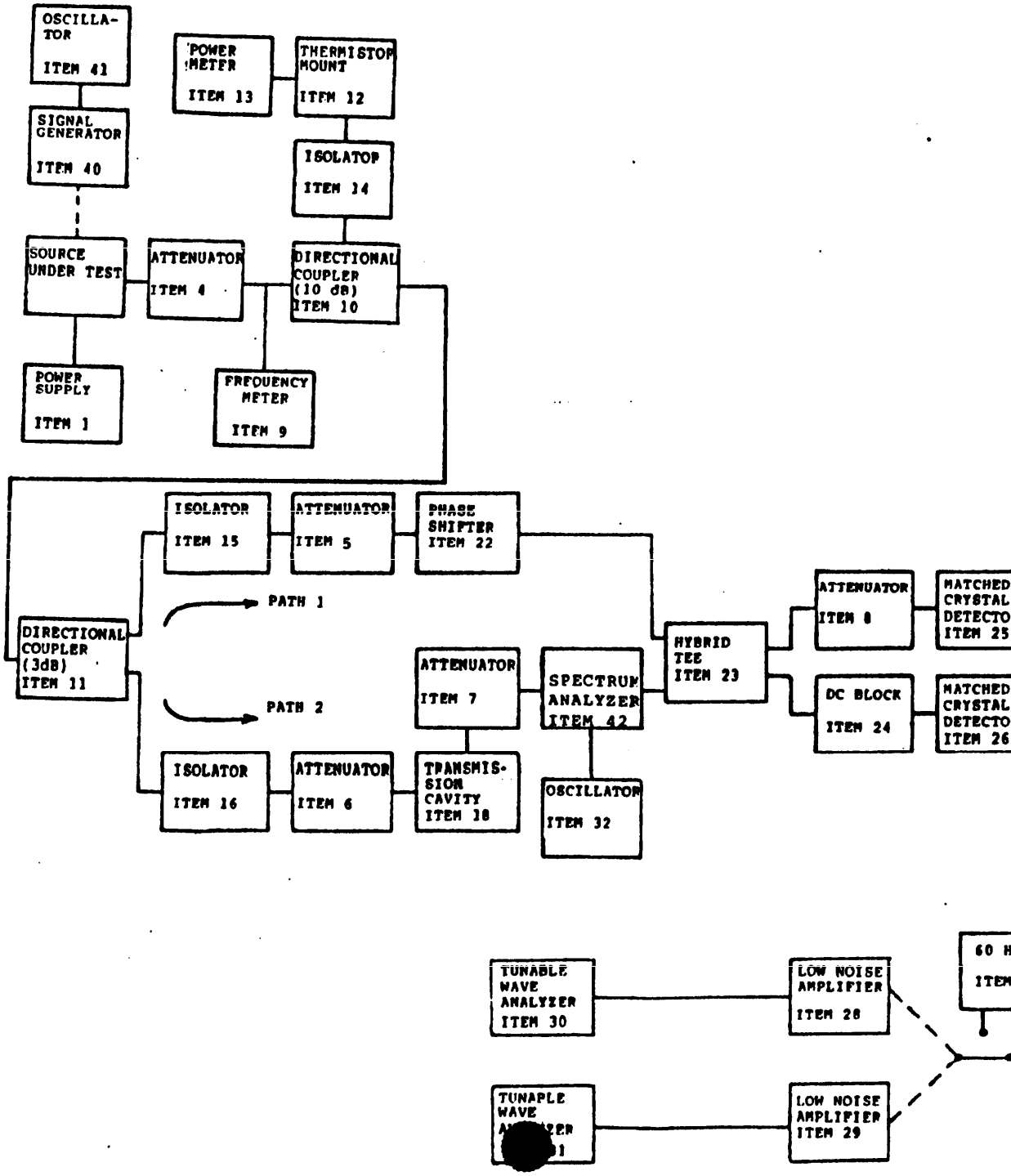


FIGURE 3. Spectral purity test set-up.

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4.7.3.2.1 Procedure. Install crystal oscillator assembly 10549175 (marked 9090 MHz), and measure the frequency using the frequency meter (item 9). Remove oscillator 10549175 and in turn, replace it with assemblies 10549176 through 10549179 and 10548224.

4.7.3.3 Output power. Assemble the equipment as shown in Figure 2 and verify compliance with paragraph 3.5.3.

4.7.3.3.1 With attenuator (Table 111, item 4) set for zero, measure power output with a termination having a VSWR of 1.05:1 or better.

4.7.3.3.2 With attenuator (Table III, item 4) set for zero, measure power output with a termination having a VSWR of 1.3:1.

4.7.3.4 Long term stability. Assemble test equipment in approved method as defined in 3.5.4 and verify compliance with 3.5.4.

4.7.3.5 Start-up time. Assembly equipment as shown in Figure 2. Verify compliance with 3.5.5.

4.7.3.5.1 Adjust the electronic counter to measure 10^{-4} second intervals Comet counter for a start trigger from power supply (Table III, item 1), turn on and stop trigger from i-f build up sensed by the matched crystal detector (Table III, item 25). Apply power to the oscillater and measure the interval from power application to rf build up.

4.7.3.6 Noise spectral density (broadband). Assemble test equipment in the approved method as defined in 3.5.6.1. A setup deemed satisfactory-for measuring spectral noise (a-m and fm) is shown in Figure 3. Verify compliance with 3.5.6.1.

Item 44 of Table III may be used as an alternate for measuring noise per 3.5.6.1. Acceptance criteria for item 44 will be in accordance with Technical Manual. When using item 4, the calibration and maintenance will be accomplished in accordance with the Technical Manual. Regardless of measurement method in event of conflict, the acceptance criteria of this specification shall prevail.

4.7.3.7 Specular outputs (narrowband). Assemble test equipment in the approved method defined in 3.5.6.1. A setup deemed satisfactory for measuring specular outputs is shown in Figure 3. Verify compliance with 3.5.6.2, using spectrum analyzer (Table III, item 42)

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4.8 System compatibility. The applicable test methods and procedure of MIL-R-50651 shall be utilized to test for the transmission characteristic, system noise figure, and subclutter visibility requirements specified in 3.5.7. The X-band source shall conform to the requirements of 3.5.7.

4.9 Burn-in. The x-band source shall be temperature cycled in test chambers capable of providing the temperature extremes and soak time specified in 3.7. During the temperature cycles the rate of temperature change within the chamber shall be 6 ± 1 degree C. per minute. A total of 6 cycles is required.

4.10. Soldering: soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.6.

5. PACKAGING

5.1 General. preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS10548182. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The X-band source is required to generate one of six signals that are used in the transmitting and receiving system of the radar. The X-band source is used in the Microwave Chassis Assembly, 10549047, a subassembly of the Receiver-Transmitter RT-860/VPS-2 .

6.2. Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
First Article Sample requirements.
c. Packaging requirements, if other than specified in Section 5.
d. Serialization requirements, if applicable.
e. Certificate of Conformance for each lot or shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

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6.3 Submission of contractor inspection equipment designs for approval. submit two copies of designs as required to: commanander, U.S. Army AMCCOM, ATTN : AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 **Drawings. Drawings listed in Section 2** of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army **Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.**

6.5 Subject term (Key word) listing.

Fire Control System
Microwave chassis assembly
Radar Set AN/VPS-2
Radar Test Set AN/TPM-22
Receiver-Transmitter RT-860/VPS-2
Vulcan Air Defence System (VADS)

6.6 **Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.**

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A786)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-48431B (AR)	29 March 1991

3. DOCUMENT TITLE

SOLID STATE X-BAND SOURCE: 10548182

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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